

A square-root counterexample to the proposed Kingman rate Conjecture 00000008482

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Abstract

The deterministic stationary subadditive family $X_{m,n} = \sqrt{n-m}$ has unit one-step costs, growth at most $n-m$, and successive increments at most one. Its limiting normalized expectation is zero, and its convergence error is exactly $n^{-1/2}$. This is not $O(\log n/n)$, refuting the proposed universal upper bound under the reading of bounded costs or bounded increments. We explicitly distinguish the alternative reading of a single bound over all interval lengths: under that reading every error is $O(1/n)$, which instead refutes the asserted logarithmic optimality. Both readings are checked in Lean 4 with Mathlib: the square-root counterexample refutes the upper bound, while uniform boundedness forces zero limit and error $o(\log n/n)$, refuting logarithmic sharpness.

1 Statement and interpretation

Problem **00000008482** states, verbatim:

Definition: The convergence error of Kingman’s subadditive ergodic theorem: the decay order of $|n^{-1}E[X_n] - \gamma|$ in n . Conjecture: The universal upper bound for bounded subadditive families is $O(\log n/n)$ (a logarithmic harmonic law); the bound cannot be improved, extremal examples being realized by discretization deviations of Sturmian balanced words (a Sturmian extremal law); and in the unbounded case the error can decay arbitrarily slowly (an arbitrary slowness law). (Kingman Sturmian extremal logarithmic decay)

We identify X_n with $X_{0,n}$ of a stationary two-index subadditive process. The conjecture does not define “bounded.” Our primary reading is bounded local costs or bounded increments, as used for subadditive processes that can accumulate costs over an interval. We require the following explicit, simultaneous restrictions:

$$0 \leq X_{m,n} \leq n - m, \quad X_{m,m+1} = 1, \quad |X_{m,n+1} - X_{m,n}| \leq 1 \quad (m \leq n).$$

The first condition bounds all normalized interval costs, and also implies that each $X_{m,n}$ is a bounded random variable. The second uniformly bounds one-step costs; the third uniformly bounds successive increments. These are substantial restrictions, imposed together on the counterexample.

There is another literal reading: one constant B bounds $|X_{m,n}|$ for every interval, independently of its length. Our example does *not* satisfy that condition, since $X_{0,n} = \sqrt{n}$ is unbounded as n varies. We do not claim otherwise. Section 4 addresses that reading directly. Under it the first conjunct is true, but the claimed optimality of the logarithmic rate is false. Thus we neither silently redefine boundedness nor claim that this particular counterexample refutes the first conjunct under every possible meaning of “bounded.”

2 The deterministic process

For integers $0 \leq m \leq n$, set

$$X_{m,n} = \sqrt{n - m}.$$

On a probability space this means the constant random variable with that value. In particular all expectations exist and $a_n = \mathbb{E}[X_{0,n}] = \sqrt{n}$. One can take a one-point probability space with the identity shift; this shift is measure preserving and ergodic. Thus the example belongs even to the ergodic setting, not merely the stationary setting. Joint stationarity holds pathwise, because

$$X_{m+k,n+k} = \sqrt{(n+k) - (m+k)} = X_{m,n}.$$

Lemma 1. For $a, b \geq 0$, $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$.

Proof. Both sides are nonnegative and $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab} \geq a + b$. Squaring therefore proves the inequality. $\square$

For $l \leq m \leq n$, put $a = m - l$, $b = n - m$. The lemma yields

$$X_{l,n} \leq X_{l,m} + X_{m,n}.$$

Nonnegativity is immediate. For an integer $d = n - m$, either $d = 0$ or $d \geq 1$, and in both cases $\sqrt{d} \leq d$. This proves the linear bound. The one-step cost is $\sqrt{1} = 1$. Monotonicity and the lemma with $b = 1$ give

$$0 \leq \sqrt{d+1} - \sqrt{d} \leq 1.$$

Hence the stated absolute increment bound holds for every interval. All negative-part integrability assumptions of Kingman's theorem are automatic, and $\mathbb{E}[X_{0,1}] = 1 < \infty$.

3 Limit and failure of the asserted rate

For every $n \geq 1$,

$$\frac{a_n}{n} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}.$$

Consequently

$$\gamma = \lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n} = 0.$$

The equality with the infimum follows because all terms are nonnegative and converge to zero. Thus the convergence error is exactly

$$\left| \frac{a_n}{n} - \gamma \right| = \frac{1}{\sqrt{n}}.$$

Theorem 2. *This error is not $O(\log n/n)$. There is no per-family big- O bound, and therefore no bound with a constant uniform over families.*

Proof. For $n \geq 2$, the ratio of the error to the proposed scale is

$$\frac{1/\sqrt{n}}{\log n/n} = \frac{\sqrt{n}}{\log n} \longrightarrow \infty.$$

For completeness the elementary logarithm estimate is enough to establish this divergence: for $x \geq 1$,

$$\log x = \int_1^x \frac{dt}{t} \leq \int_1^x t^{-3/4} dt = 4(x^{1/4} - 1) \leq 4x^{1/4}.$$

It follows that $\log x/\sqrt{x} \rightarrow 0$ and the displayed reciprocal diverges. Equivalently, a purported eventual inequality $\sqrt{n} \leq C \log n$ is incompatible with $\log n = o(\sqrt{n})$. The latter is the form used in Lean. Multiplying the hypothetical big-O bound by n gives $\sqrt{n} = O(\log n)$; composing with $\log n = o(\sqrt{n})$ gives $\sqrt{n} = o(\sqrt{n})$, impossible since $\sqrt{n} > 0$ eventually. This argument allows both the constant and the starting index to depend on the family, so it disproves the weaker per-family reading as well as the uniform reading. $\square$

The original conjecture is a conjunction. Under the bounded-costs reading, Theorem 2 refutes its first conjunct, so no separate disproof of the Sturmian assertion or the arbitrary-slowness assertion is required. We make no claim here that the arbitrary-slowness assertion is false.

4 Robustness to the reading of 'bounded'

Suppose instead that a single $B < \infty$ satisfies $|X_{m,n}(\omega)| \leq B$ for all intervals almost surely. Then $|\mathbb{E}[X_{0,n}]| \leq B$, so $a_n/n \rightarrow 0$, and

$$\gamma = 0, \quad |a_n/n - \gamma| \leq B/n.$$

The same reasoning applies if the single bound is imposed directly on the expectations. Since $1/n = o(\log n/n)$, the logarithmic order is strictly improvable throughout this class. In particular no example in this class, Sturmian or otherwise, can attain a non-improvable $\log n/n$ order.

The $1/n$ order itself can be attained. On a one-point probability space take $Y_{m,n} = 1$ when $m < n$ and $Y_{m,m} = 0$. It is stationary and globally bounded by one. For $l \leq m \leq n$, if $l = n$ the subadditivity inequality is zero on both sides. If $l < n$, at least one of $l < m$ or $m < n$ holds, so the right side is at least one, proving subadditivity. Now $\mathbb{E}[Y_{0,n}] = 1$ for $n \geq 1$ and the error is exactly $1/n$. The uniform-boundedness argument is now also formalized. In Lean, `UniformlyBounded X` means exactly

$$\exists C \in \mathbb{R}, \quad \forall m, n \in \mathbb{N}, \quad |X_{m,n}| \leq C.$$

The theorem `uniformlyBounded_limit` proves both convergence of $X_{0,n}/n$ to zero and `gamma X = 0`; `uniformlyBounded_error_isLittleO` proves

$$|X_{0,n}/n - \gamma X| = o(\log n/n).$$

These results require only uniform boundedness. The little-o proof uses $X_{0,n} = O(1)$ and $1 = o(\log n)$, then multiplies both comparison functions by $1/n$ and takes the absolute value. Thus the first upper-bound clause is true under this reading, but its claimed sharpness is false.

For this reading the admissibility predicate `UniformlyBoundedStationarySubadditive` contains exactly translation stationarity, subadditivity on ordered triples, and `UniformlyBounded`. It imposes no nonnegativity, normalization, unit one-step cost, or separate increment restriction. The sharpness clause `SharpnessClauseUniform` asserts the existence of such a family whose error is *not* $o(\log n/n)$. The checked theorem `sharpness_false_uniform` rules out every such witness. The combined theorem is

```
theorem conjecture_00000008482_false_both_readings :
  Not UniversalUpperBound /\ Not SharpnessClauseUniform
```

Under the linear-growth or bounded-increment reading the first conjunct fails, witnessed by the square-root family. Under the uniformly bounded reading the second conjunct fails for every admissible family. Either way the original conjunction is false. The $1/n$ witness just described is a written example; the universal limit and little-o conclusions, sharpness refutation, and combined result are all Lean theorems.

5 Formal statement and semantic audit

The project pins Lean `leanprover/lean4:v4.33.0` and Mathlib `v4.33.0`. The Mathlib revision is: `db584cd6d46c92f209a44c0f1c829460d327499d`

The single source file is `lean4/KingmanCounterexample.lean`. All declarations are in namespace `KingmanCounterexample`. Family abbreviates $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$. The value of `sqrtFamily m n` is `Real.sqrt (n - m : Nat)`. Only ordered intervals are relevant; natural subtraction simply supplies harmless values elsewhere.

The predicate `BoundedStationarySubadditive X` has exactly five fields: subadditivity on ordered triples, translation stationarity, the nonnegative linear bound, unit one-step costs, and the absolute successive-increment bound. Requiring exact unit costs narrows the class further, making the negative result stronger: the broader conjectured universal bound would imply this restricted universal bound.

`gamma X` is defined as

```
limUnder atTop (fun n : Nat => X 0 n / (n : Real))
```

This is Mathlib’s limit operator. We do not presume convergence for an arbitrary function: the universal clause explicitly requires convergence to this value. Our witness satisfies this extra premise, as proved by `sqrtFamily_limit` and `sqrtFamily_gamma`. The universal clause is the following actual Lean definition (ASCII spellings of mathematical symbols are used here):

```
def UniversalUpperBound : Prop :=
  forall X : Family, BoundedStationarySubadditive X ->
    Tendsto (fun n : Nat => X 0 n / (n : Real))
      atTop (nhds (gamma X)) ->
    Asymptotics.IsBigO atTop
      (fun n : Nat => abs (X 0 n / (n : Real) - gamma X))
      (fun n => Real.log (n : Real) / (n : Real))
```

```
theorem conjecture_00000008482_false :
  Not UniversalUpperBound
```

The displayed definition is a typographical transcription of the source’s Unicode notation, with the same quantifiers and arguments. Mathlib’s `IsBigO` has its usual meaning: an eventual norm bound with some real constant. The filter is the natural-number `atTop`. Therefore this is a genuine asymptotic assertion on the full sequence, not a finite sample or selected subsequence.

The supporting checked theorems establish the following:

- `sqrtFamily_hypotheses`: all five family conditions.
- `sqrtFamily_error`: the error is the reciprocal square root.
- `sqrtFamily_limit` and `sqrtFamily_gamma`: the limit exists and equals zero.
- `sqrtFamily_infimum`: the infimum over lengths $n + 1$ is zero, covering every positive length without admitting length zero.
- `sqrtFamily_not_bigO`: the error fails the asserted big-O bound.
- `conjecture_00000008482_false`: the restricted universal clause is false, using that explicit witness.
- `uniformlyBounded_limit`: uniform boundedness forces convergence and `gamma X = 0`.

- `uniformlyBounded_error_isLittle0`: every such error is $o(\log n/n)$.
- `sharpness_false_uniform`: the uniformly bounded sharpness clause is false with exactly the three relevant admissibility conditions.
- `conjecture_00000008482_false_both_readings`: the two refutations hold together.

At length zero Lean uses totalized division by zero, giving zero; this does not affect any `atTop` assertion or the positive-length infimum. The deterministic formulation needs no expectation machinery: constant random variables on any probability space have exactly these expectations. The probabilistic embedding and one-point ergodicity explanation above are elementary written arguments.

6 Verification and attribution

The Lean compilation succeeded with Lean and Mathlib v4.33.0; `verification.txt` records the compiler output and axiom audit. The audited main and supporting theorems use only `propext`, `Classical.choice`, and `Quot.sound`. There are no added axioms or incomplete proofs. The asymptotic step uses Mathlib’s proved theorem `isLittle0_log_rpow_atTop` with exponent $1/2$, composed with natural-number inclusion, and a proved big-O/little-o incompatibility.

`verify.py` is an independent Python standard-library sanity check of subadditivity on finite ordered triples, stationarity under shifts, the bounds, the exact error formula numerically, and growing ratios on geometrically separated indices. Floating-point checks are explicitly supplementary; none is used as a proof of an infinite assertion. To reproduce in a fresh environment:

```
cd lean4
lake exe cache get && lake build
cd ..
python3 verify.py
pdflatex report.tex
pdflatex report.tex
```

No external project code is vendored. The source conjecture is available in the competition repository at `conjectures/00000008482.md`.

Submitted by AlyciaBHZ on behalf of the Omega Institute (trureturing project, <https://github.com/the-omega-institute/trureturing>).